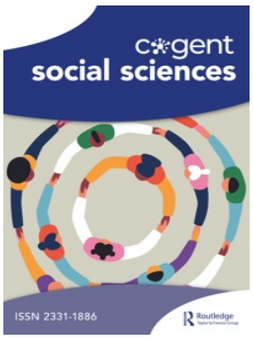

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Unraveling the conceptual ambiguity of digital human resource management: a state-of-the-art review

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ABSTRACT

Digitalization is the fundamental aspect of the technological revolution that has introduced organizations to disruptive change. Not only external, but digitalization has also transformed all the internal mechanisms of organizations. The digital transformation of human resource management has been identified as a novel phenomenon. Therefore, during the past three decades, academicians have paid remarkable attention to and investigated the integration of technology with human resource management. Researchers strived to define this phenomenon in various ways and represent it with multiple terms. Interchangeable terms and different definitions have created ambiguity, and as a result, literature still lacks conceptual clarity. To address this gap, this study conducted a state-of-the-art review by following the six – step approach. The objective of this review is to clarify the conceptual ambiguity. This study employed Scopus, Web of Science and Google Scholar to identify relevant studies. A total of 1,274 articles were reviewed until finally 19 articles were identified as the subject of research. Firstly, this study presents the chronological development of the concept. Secondly, this study has analyzed the definitions of digital human resource management and its related concepts to identify the ambiguity that is misleading the literature in different directions. Through analysis this study explained the misunderstandings among different terms and concepts. The findings of this study have shown that digital HRM is a process of integrating human activities with advanced technology. Lastly, this study suggested an area for future research to further explore this evolving field.

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Introduction

The digital revolution across industries has entirely changed the work processes within organizations (Imamov & Semenikhina, 2021). Technological advancement is a principal driver of change within an organization (Dosi & Nelson, 2010; Rana, 2019). It becomes essential for most organizations to efficiently integrate their systems with technology not only to survive but also to stay resilient and thrive in a highly competitive environment (He et al., 2021; Luftman et al., 1993). It can only be achieved through transforming the existing processes into digital (Fosso Wamba et al., 2015; White, 2012). Digital transformation is the embedment of disruptive technologies into existing processes to improve efficiency and enhance the engagement of stakeholders. (Ebert & Duarte, 2018; Vial, 2021). Digital transformation introduced the operating environment of the organization with a disruptive change (Parviainen et al., 2017). which ultimately triggered significant changes in every domain of business and society. The domain of Human resource management is no exception (Zavyalova et al., 2022). Digital transformation has taken traditional HRM into the new realm of technology (Bondarouk et al., 2017). Where unimaginable practices such as approaching and recruiting diverse candidates are possible within the shortest possible

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time and lowest budget. Such as in the past decades training the same course for all staff at a time was hectic and almost impossible due to heavy resource investment. Presently e-learning, a core concern of digital HRM practices, made it possible (Bondarouk et al., 2017). Where digital revolution has advanced the process of human resource management it has also advanced the concepts of human resource management i.e. Digital HRM (da Silva et al., 2022; Jani et al., 2021).

Researchers are inquisitive about the digital perspective of HRM because it addresses much more than just the digital shift. At present, academician are paying high attention to the concept of digital HRM and its related concepts including digitization, digitalization, digital transformation, and digital disruption of HRM (Sima et al., 2020; Strohmeier, 2020). Several researchers contributed to establishing the understanding of the concept of digital HRM. The literature still lacks unanimity over a single definition. Researchers have described the concept of digital HRM interchangeably with its related concepts. Whereas some researchers linked the concept of digital HRM with past concepts described in the literature. However, the use of these concepts is proliferating, heterogeneous, and implicit (Strohmeier, 2020). This conflicting interpretation causes misunderstanding among stakeholders and makes it challenging to operationalize the concept of digital HRM (Bondarouk & Ruël, 2009; Zavyalova et al., 2022). These concepts lack conceptual clarity which requires further research (Thite, 2022). For instance, manager and employees do not correctly understand the concept of digital HRM. Consequently, misunderstand the related benefits of digital HRM and perceive it as threat to job security. As a result, resistance may hinder the successful adoption of digital HRM. Further, misinterpretation of HR analytics due to misunderstanding of digital HRM concept may affect the decision making. It is crucial for HR professionals, senior management and IT experts to have clear understanding digital HRM concept for its effective implementation.

The concept of digital HRM is under limelight in the academic world. Researchers are paying more attention to understanding the concept of digital HRM and striving to precisely define it (Bondarouk et al., 2017; Strohmeier, 2020; Thite, 2018, 2022). Strohmeier (2020) argued that the concept of digital HRM and its related concepts i.e. digitalization, digital transformation, digitization, and digital disruption are used in an implicit, heterogeneous, and proliferating manner. Various researchers strive to describe this concept, but it still lacks clarity concerning its related concepts. The boundaries that could help to clarify it remained blurred (Thite, 2022). Conceptual clarification and development of digital HRM concept is a major challenge in the research domain (Bondarouk & Ruël, 2009; Mitlacher, 2012; Ruël & Bondarouk, 2014; Strohmeier, 2020; Thite, 2022). This challenge calls for an immediate solution. For the attainment of this research problem first, it is necessary to have a clear understanding of the phenomenon under study. A slight switch in concepts may lead research in a different direction (Bondarouk & Ruël, 2009). The increasing importance of this concept calls for formal categorization in literature. To address the argument about conceptual clarity and developing a new theoretical perspective of digital HRM, it is essential to know the chronological development of the concept. The research question of this study aims to address mentioned as follows:

Research question: What is digital HRM and How did the understanding of the concept of digital HRM evolve?

Aforementioned research question aims to explore the evolution of the concept of Digital HRM and clarify the definition. It seeks to understand the essence of digital HRM and track its conceptual development over time. This study highlights the multiple interpretations of the concept in literature. By exploring the chronological development of digital HRM concept, researchers aim to get insights into its foundational principles, emergent themes, and evolving perspectives within the field of HRM and technology integration. To address this research question, it is required to conduct an in-depth analysis of the existing body of literature. For this purpose, the present study opts to conduct a state-of-the-art review. In the state-of-the-art review author(s) collect and the analyze the chronological development of the body of literature for a certain phenomenon, critically review the evolvement of the concept, and address the reason for the development of the concept in a certain and direct for future research (Barry et al., 2022; Grant & Booth, 2009). Further, scientific disciplines must conduct a literature review grounded on clearly reported thorough, and rigorous methodology (Snyder, 2019). It ensures the quality and trustworthiness of research. As mentioned earlier Strohmeier (2020) criticized the field's understanding of proliferating utilization and the heterogenous application of digital HRM concepts. The author's arguments are not supported by a thorough and rigorous review of the literature. Which raises a question

about the validity of the argument. This concern is also supported by the publication of Thite (2022) which calls for further research on conceptual clarity. The objective of this research is to clarify the conceptual ambiguity by providing a chronological overview of the present state of knowledge of digital HR. Further, this study develops understanding about the evolution of digital HRM concept and suggests a direction for future research.

Methodology

The literature does not report any established methodology to conduct the SotA review. Therefore present study followed the recently developed six-steps methodology of Barry et al. (2022) as shown in Figure 1. This six – step approach ensured the methodological rigor, enhancing reliability and quality. It serves as a roadmap for conducting a comprehensive and credible synthesis of existing literature.

Step 1: Determine the field of inquiry

Multiple definitions of a single concept and its interchangeable utilization with multiple terminologies by researchers created a big confusion in the existing literature. Due to this most researchers misunderstood the concept and operationalized the concept based on their subjective understanding. The present review aimed to synthesize the existing established definitions of digital HRM to establish an understating of digital HRM. And to explore how the concept of digital HRM evolved. The initial research question of this review is as follows

R1: What is Digital HRM and How has the concept of digital HRM evolved?

As mentioned earlier the concept of digital HRM is interchangeably used with multiple terms and researchers also denoted with past concepts. Therefore, to address the research question it is necessary to identify all the terms with which the concept of digital HRM has interchangeably used and also search for the past concepts with which digital HRM has been linked. For this purpose, the present study has employed the VOS viewer software (van Eck & Waltman, 2011). Through a text co – co-occurrence map this study strived to find the terms and concepts related to digital HRM. This practice will further assist in determining the search strategy and timeframe to develop a pool of all relevant studies for analysis.

Scopus and Web of Science are widely utilized databases for bibliometric analysis. VOS viewer allows the engagement of one database at a time therefore this study opts for the Scopus database. Scopus is the largest multidisciplinary database containing peer-reviewed literature in social science research (Chadegani et al., 2013; Donthu et al., 2020). Scopus database was searched with 'digital' AND 'HRM' OR 'Human Resource Management' to conduct the keywords co–occurrence analysis. An initial total of 3,369 keywords were identified, after removing the keywords with no meaning and frequency lesser than 3, a total of 337 keywords remained. 100 key terms were not connecting to any other term therefore the same have been omitted from the map. The map consists of 237 connected terms. The text co–occurrence map (see Figure 2) highlights widely published key terms co – occurred with digital HRM and the era in which specific term has been vastly focused. Stating the year does not mean that the specified terms have never been used before or studies stopped utilizing them afterward. Figure 2 reflects the key terms that appeared with and linked with digital HRM i.e. Human Resource Information System (HRIS), Electronic Human Resource Management (e–HRM), and Digital Transformation. The studies used the term digital HRM and also cited the studies to relate digital HRM with past concepts or interchangeably used digital HRM with past concepts.

From Figure 2 it can be seen that until 2012 the major focus of the research was on the term HRIS. Researchers started to operationalize the terms e-hrm, and electronic hrm from 2012 onward. The major focus on the term e-hrm was from 2014 to 2018. In 2018 studies slightly focused on the term Management

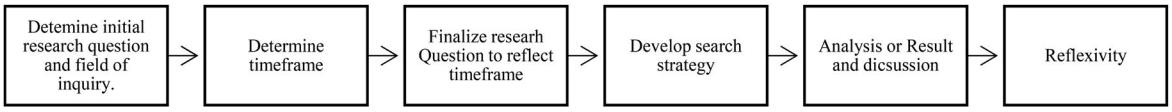

Figure 1. Six – steps of SotA.

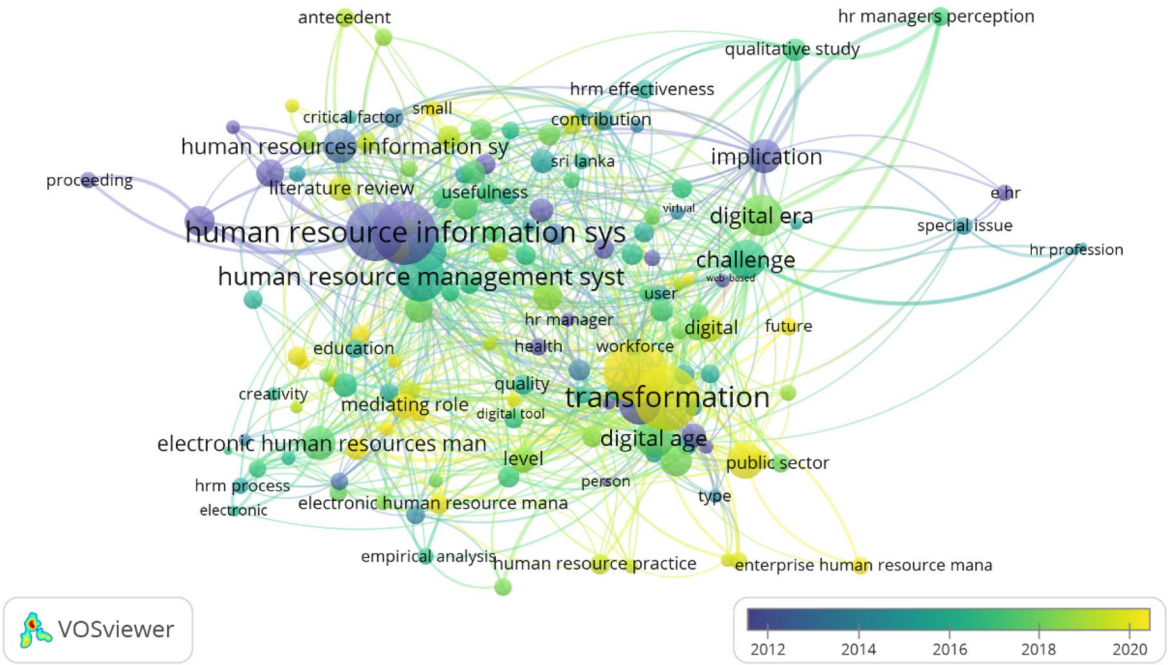

Figure 2. Text co – occurrence map.

information system which is the interchangeable term HRIS. Although the term digital transformation of HRM evolved in 2017 it has gotten academic attention in 2020.

Step 2: Determine timeframe

The structure of HRM has evolved over the past three decades due to technological advancement. Understanding of technology integration in the present era of advanced technology is different from the earlier developed understating. To gauge the difference between the two ends of understanding it is necessary to apprehend the inception of the concept. And chronologically review the evolution. From text co – occurrence analysis this study has identified the initial search term i.e. Human Resource Information System (HRIS). The search was limited to articles published in the English language and peer-reviewed journals published between 1970 and 2023 to include all definitions described over time. The base year 1970 was decided because since the mid-80 industries have started to integrate information systems in HR functions (Von Glinow & Teagarden, 1988). The first study was published in 1970 that introduced the concept of information systems in HR function (Wilkens, 1971). Therefore 1970 is the sentinel point considered the beginning of the digital HRM concept.

Step 3: Finalize the research question to reflect the timeframe

At this stage researcher(s) are likely to revise the research question based on the description of stage 2 (Barry et al., 2022). For the present study, this is not the case because the understanding of the concept has already been described in the introduction section. To avoid repetition the same has not been reported in stage 2.

Step 4: Develop a search strategy and collect relevant studies

To develop the initial pool of articles defining the concept of digital HRM and its related concepts that include digitalization, digitization, and digital transformation. Through step 1 and the literature search, it has been identified that digital HRM is interchangeably used with e-HRM. Whereas e-HRM is used interchangeably with virtual HRM, HR intranet, web-based HR, computer-based human resource

management systems (CHRIS), and HR portals (Wen, 2013). Therefore, these interchanged terms were considered while developing the search strategy. The database was searched with the following search strings:

'Digital*' OR 'Electronic' OR 'E' OR 'Web-based' OR 'computer-based' OR 'Digitization' OR 'virtual' AND 'HRM' OR 'Human Resource Management'

'Human resource information system' OR 'HRIS' OR 'Human resource Management system'.

Quality assessment was beyond the scope of SotA review, still, this study focused on the studies published in well-reputed peer-reviewed journals. At first Scopus database was searched with the above-defined search strategy. A total of 544 studies were identified including empirical studies, book chapters, and conceptual and review papers. Secondly, the Web of Science database was searched with the same strategy. A total of 380 records were identified. Further to ensure the comprehensive retrieval search engine, Google Scholar, was also browsed. After removing 166 duplications, 184 additional records were retrieved. Given below chart reflects the concept-wise search outcomes (see Figure 3).

All articles were thoroughly reviewed and separated into two categories. In the first category articles presenting the definition of concepts, reporting the conceptual clarification, and review articles were included. In the second category, all the empirical studies were included. The focus of this literature is synthesizing the established definitions therefore only book chapters, conceptual and review papers defining and explaining concept were retained for analysis. Out of 1,108 records, only 168 records matched the selection criteria.

Data extraction

As shown in Figure 4, the number of publications on digital HRM and related concepts has increased during the time (1971–2023). The concept of HRIS has been operational since 1971, the time multiple interchangeable concepts have been introduced but the concept of HRIS in research is still operational. The concept of electronic HRM has also gotten remarkable attention. A significant volume of publications on the concept of electronic HRM (321) has been identified over time. Researchers have also started to explore the term digital HRM since 2018 publications on digital HRM took a sudden boom. Whereas the concept of virtual HRM, Web–based HRM, digitization, and digitalization of HRM are not much famous. The important point that needs attention over here is the definitions researchers operationalize to define these concepts.

This study extracted three kinds of information from the shortlisted papers. First the descriptive data of the article that includes the year of publication, and type of paper (book chapter, conceptual paper, or review paper). Empirical studies are already separated earlier. Second the purpose/objective of the article. Lastly, each article was examined and it was identified what definition has been operationalized in the article. After thoroughly reviewing the selected 168 records only 19 studies were identified that established the definition at different times based on concrete conceptualization. Rest 149 studies cited

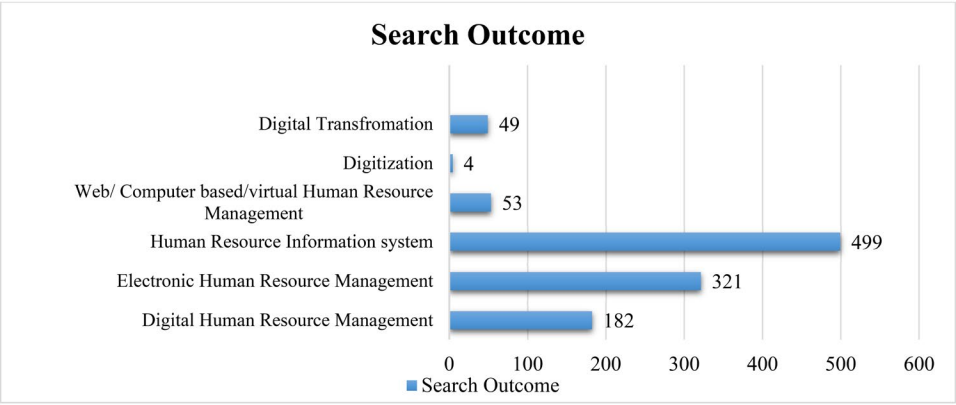

Figure 3. Search Outcome.

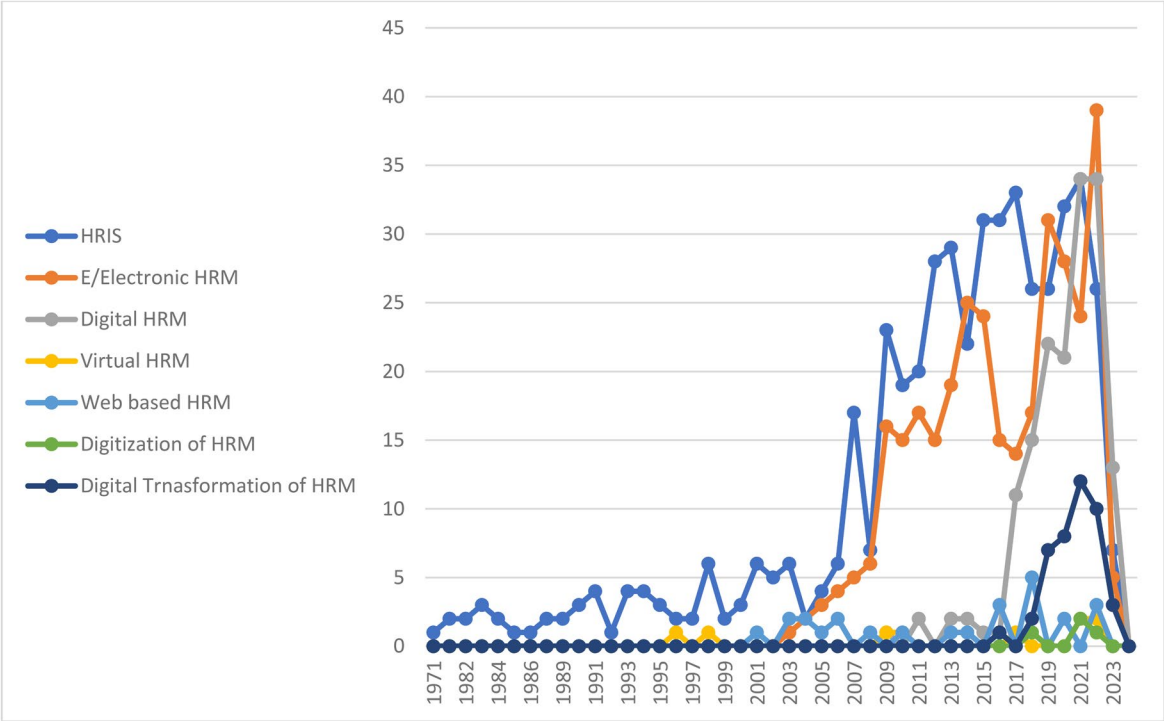

Figure 4. Publication trend.

and summarized these 19 articles, and no concept development has been reported. Most of the studies did not differentiate these terms and considered all these terminologies as one. Only 19 studies provided the corpus for analysis. Since the objective of the review is to extract and analyze the definitions therefore for no discrepancy has been found during data extraction. The PRISMA flowchart (Page et al., 2021) reflects the record selection process (See Figure 5)

Key definition-related resources identified in Step 2 and data extracted in Step 3 were inductively analyzed to identify the underlying explanation of concepts and components of definitions. Step 3 revealed that the concept of 'Digital HRM' was described in 6 different ways across the 1,108 articles. Out of 168 studies claiming conceptual clarification, concept development, and systematic and narrative review, only 19 articles clearly define the concept. remaining studies lack supporting information, and these studies do not guide how the concept of digital HRM evolved. These studies were just narrative reviews providing inconclusive results.

Step 5: Result and discussion

This study used a content analysis technique to analyze the extracted data. Across the 19 studies, three major concepts were identified and defined by different authors. At first, the definitions of these concepts was separately analyzed and identified how these definitions are different from each other. Secondly, these three concepts were compared with each other to verify if these concepts are related to each other and progressed over time. this study has extracted key definitions and derived the themes. Table 1 represents the key definitions of HRIS concept and extracted themes.

Human resource information system (HRIS)

The world wide web has been released to the general public in the early 90s but it was not as advanced as in today's world. Until the late 90s utilization of information technology was generally limited to the utilization of software and applications for storing and analyzing data. This study chronologically reviewed all the related studies defining the related concepts of digital HRM and its related concepts. Until the mid-90s authors described the concept according to the available technological advancements. Authors denote the utilization of technology to perform HR functions as a Human Resource Information System

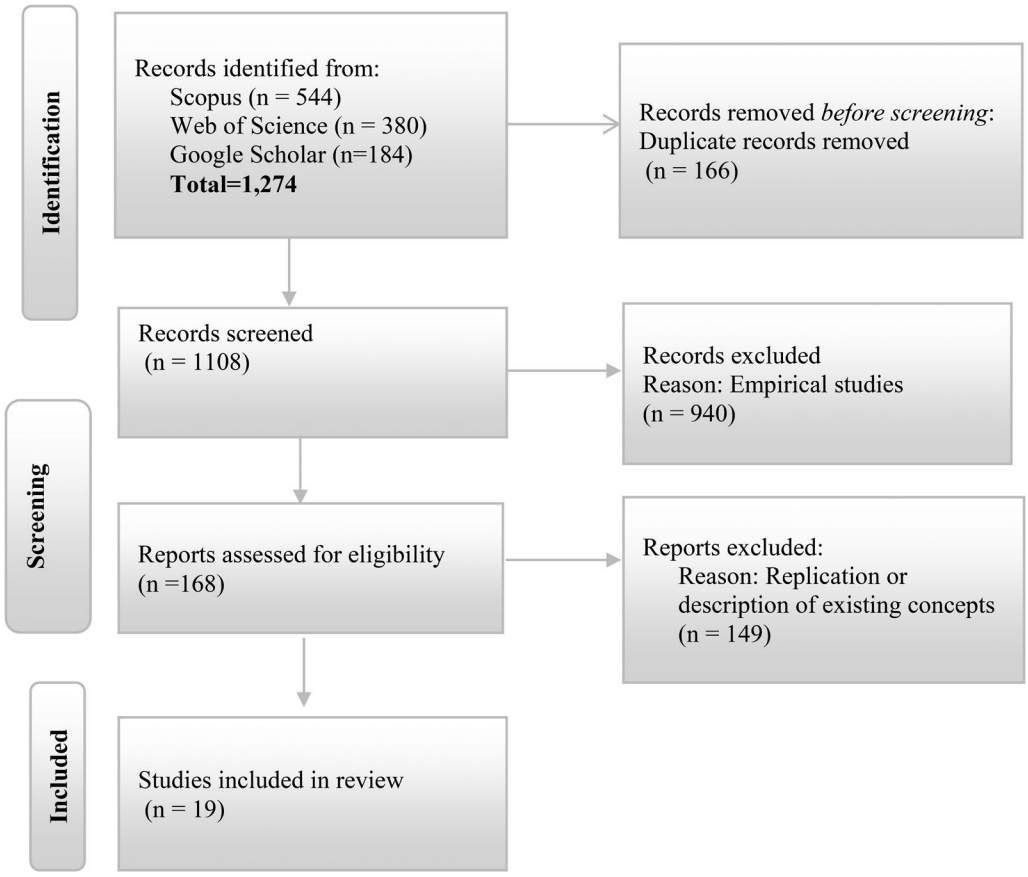

Figure 5. PRISMA Process Flow.

Table 1. Key definitions of HRIS concepts and extracted themes.

Author	Key definition	Themes
Wilkens (1971)	A cybernetic system that provides for the collection and dissemination of human resource information system	Cybernetic system, collection, and dissemination, human resource information.
Anthony (1977)	HRIS is an information system that facilitates HR managers to maximize human resource decisions. Linked to electronic data processing facilities, the system can easily be accessed, updated, and used for analysis.	Information system, data processing, data analysis, decision making
Hyde and Shafritz (1977)	HRIS is an integrated personnel management information system that produces specific reports for managers and reports them as decisional options.	Information system, personnel management, data processing, decision making.
Walker (1982)	HRIS is a systematic procedure for collecting, storing, maintaining, retrieving, and validating the data needed by an organization for its human resources, personnel activities, and organization unit characteristics.	Systematic procedure, data management, personnel management.
DeSanctis (1986)	Specialized information systems within the traditional functional areas of the organization are designed to support the planning, administration, decision-making, and control activities of human resource management.	Information system, human resource management, decision making.
Jesser (1990)	The system used to acquire, store, manipulate, analyze, retrieve, and distribute pertinent information regarding an organization's human resources	Data management, data analysis.
Broderick and Boudreau (1992)	The composite of databases, computer applications, and hardware and software necessary to collect/record, store, manage, deliver, present, and manipulate data for human resources.	Computer-based system, data management, data analysis, human resource management.
Dilu et al. (2017)	HRIS is a database system and an efficient catalyst for linking ICT (information and communication technologies) and HR management and providing essential support for HR management	Databased system, human resource management.

(HRIS). Wilkens (1971) is considered the first peer-reviewed scientific article that used the term 'Human resource information system' to describe the advancement in the HRM concept. The author just introduced the term HRIS and briefly defined it. Author described HRIS as a cybernetic system, a complex framework that help organizations in managing, controlling and disseminating information related to human resource. This research lacks a further explanation of the concept, how this system work within the organization. Later on, researchers operationalize the term HRIS to introduce the concept of

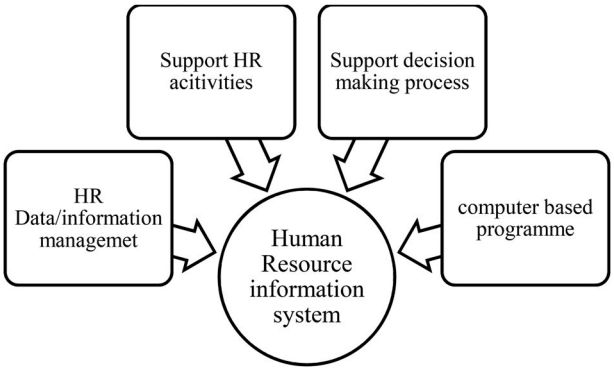

Figure 6. Conceptual Map of HRIS.

technology-based HRM. They used the term 'Human Resource Information Systems' (HRIS) and highlighted the incorporation of technology for personnel management. The authors explained it as an information management system that automatically manages the basic information related to HR activities and supports managers in the decision on recruitment, placement, training, classification, and resource allocation (Anthony, 1977; Hyde & Shafritz, 1977). These authors strived to define the concept of HRIS in a novel way and explained its implications. The point that needs attention is after the first publication none of the next two authors cited the first publication neither of these two authors has been cited by future researchers. The studies published after 1986 reported DeSanctis (1986) as the first peer-reviewed article that defined the concept of HRIS. The focus of the studies was to describe the basic issue of HRIS related to the planning and development of HR functions and provide comprehensive guidelines (Lederer, 1984a, 1984b; Nikkel, 1985; Simon, 1983).

Authors described HRIS as a computer-based system or software that possesses the capability to manage HR data and help managers in the decision-making process (Broderick & Boudreau, 1992; DeSanctis, 1986; Dilu et al., 2017; Jesser, 1990). None of the above-mentioned authors have made any notable contribution to extending the concept of HRIS. All authors endorsed the description of Anthony (1977) and Hyde and Shafritz (1977). They all defined the HRIS in the same manner only the arrangement of words and induction of advanced technological terms distinguish these definitions. Such as (Walker, 1982) defined HRIS as 'A systematic procedure for collecting, storing, maintaining, retrieving, and validating the data needed by an organization for its human resources, personnel activities, and organization unit characteristics.' After 3 decades Dilu et al. (2017) defined HRIS as 'A database system and an efficient catalyst for linking ICT (information and communication technologies) and HR management and providing essential support for HR management'. They all describe HRIS as the provision of technical support in performing HR activities and managing HR data for analysis and decision-making. This study thoroughly analyzed the key definitions and extracted themes Through analysis of key definitions and the first explanation of HRIS, it has been identified that HRIS is not a theoretical concept but it is '*A socio-technical*[1] *program that helps managers in performing HR activities and efficiently managing human resources to assist stakeholders*'. Figure 6 represents that conceptual map of HRIS based on the extracted themes.

Electronic/e-HRM

Above Table 2 contains the key definitions of E-HRM explained over time and themes derived from them. Lepak and Snell (1998) used a different term 'virtual HR' to describe a 'network-based structure built on partnerships and mediated by information technologies to help the organization acquire, develop, and deploy intellectual capital'. They extended the existing concept of HRIS by embedding the use of the internet with HRIS to support and perform HRM. Lednick-Hall and Moritz (2003) for the first time used the term Electronic-HRM to extend the existing concept of virtual HRM and define it as 'conducting HR transactions using the Internet or intranet'. Author introduced the induction of internet within the existing processes of human resource management to enhance the efficiency. Later on, researchers operationalized the term e-HRM as a substitute for HRIS or virtual HRM (Wirtky et al., 2016; Ziebell et al., 2019). By comparing the definitions of these two terms it has been identified that HRIS was described as a

Table 2. Key definitions of Electronic HRM concept and extracted themes.

Author	Key definition	Themes
Lepak and Snell (1998)	The term 'virtual HR' is used to describe a 'network-based structure built on partnerships and mediated by information technologies to help the organization acquire, develop, and deploy intellectual capital.	Information technology, human resource management
Lengnick-Hall, and Moritz (2003)	conducting HR transactions using the Internet or intranet	Use of Internet, HR transaction
Ruël and Bondarouk (2014)	A way of implementing HRM strategies, policies, and practices in organizations through the conscious and direct support of and/or with the full use of channels based on web technologies.	Usage of web-based technology, management of HR activities
Strohmeier (2007)	Planning, implementation, and application of information technology for both networking and supporting at least two individual or collective actors in their shared performing of HR activities.	Incorporation of web – based information technology, human resource management, HR activities.
Bondarouk and Ruël (2009)	All possible integration mechanisms and contents between HRM and Information Technologies aim at creating value within and across organizations for targeted employees and management.	Integration of information technology, human resource management
Marler and Parry (2016)	Configurations of computer hardware, software, and electronic networking resources that enable intended or actual HRM activities (e.g. policies, practices, and services) through coordinating and controlling individual and group-level data capture and information creation and communication within and across organizational boundaries.	Configuration of information technology, HRM activities, data management, electronic networking resources.
Florkowski (2018)	The IT applications, databases, software, and equipment used for collecting, sorting, analyzing, and delivering the information for managing the HR.	Utilization of IT applications, data management,
(Berber et al., 2018)	Electronic Human Resource Management (e-HRM) is a concept that involves the use of Web-based technologies for providing services regarding human resource management in the organization, and to access a wider range of the organization's stakeholders – starting from the HRM department, via managers to employees.	Use of web–based technology, managing information, HR activities.

computer-based system to support and perform HR activities. Whereas e-HRM is defined as the process of integrating web-based technology to carry out HR activities and access a wide range of stakeholders. These are related concepts but cannot be used interchangeably.

From the above-cited definitions of e-HRM, it can be observed that none of the above authors deviated from the two identified commonalities i.e. connecting HRM with advanced web-based technology and aiming to induce the efficacy of HR functions. Some authors added access to stakeholders. The authors conceptualize the term e-HRM differently by extending the HR functions. Such as Ruël and Bondarouk (2014), defined it as 'A way of implementing HRM strategies, policies, and practices in organizations through the conscious and direct support of and/or with the full use of channels based on web technologies'. Bondarouk and Ruël (2009) criticized the previously developed concept of e-HRM that, these definitions are silent about the network of people in the organization. Later on, the authors considered this critique while redefining the concept of e-HRM. Such as Berber et al. (2018) defined it as 'Electronic Human Resource Management (e-HRM) is a concept that involves the use of Web-based technologies for providing services regarding human resource management in the organization, and to access a wider range of the organization's stakeholders – starting from the HRM department, via managers to employees'.. From the analysis, it has been identified that the authors did not conceptualize e-HRM in a contradicting manner. Every time authors conceptualize e-HRM by inducting innovative HR functions and using advanced fancy terms for technology. For example, Marler and Parry (2016) stated, configurations of computer hardware, software, and electronic networking resources. It can be seen that the authors tried to define the concept of e-HRM attractively by using the highlighted advanced technological terms. E-HRM is not restricted to the usage of web-based technology in HRM functions, but it is a wider concept that can be defined as, '*the assimilation of HRM activities and information with advanced web-based information technology to access and assist stakeholders efficiently*'. Figure 7 represents that conceptual map of E-HRM based on the extracted themes.

Digital human resource management

Out of 17 publications, only 3 attempted to define the concept of digital HRM and clearly described that the term 'digital' is interchangeably used with the term 'electronic'. For example, Thite (2018) defined it as 'Digital or electronic HRM deals with the nature, role, and contribution of technology in strategically managing talent in a digital world. It incorporates social, mobile, analytics, cloud (SMAC), and other emerging technologies for

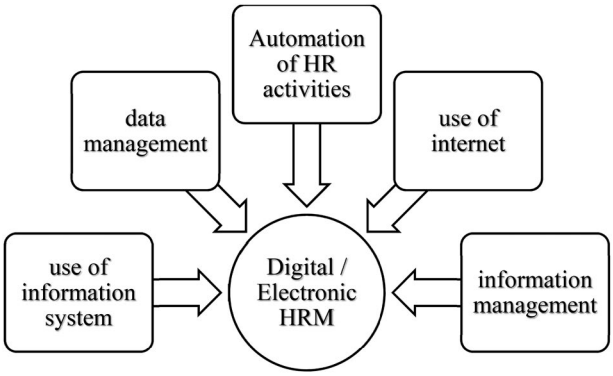

Figure 7. Conceptual Map of E-HRM.

Table 3. Key definition of Digital HRM concept and related themes.

Author (s)	Key definitions	Themes
S & Aggarwal (2017)	The usage of technology in the planning and implementation of various HR activities.	Usage of technology, HR activities.
Thite (2018)	Digital or electronic HRM deals with the nature, role, and contribution of technology in strategically managing talent in a digital world. It incorporates social, mobile, analytics, cloud (SMAC), and other emerging technologies for the efficient and effective delivery of HR services	Contribution of technology, strategic talent management, and HR services.
Strohmeier (2020).	digital HRM is best understood as an evolutionary advancement of previous conceptualizations of technology-based HRM	–

the efficient and effective delivery of HR'. In this definition author used the term 'digital' interchangeably with 'electronic' and like other definitions author emphasized the induction of advanced technology. From 2017 until 2023 this study identified 149 publications used the term 'digital', authors used it interchangeably with 'electronics'. The authors of these studies are less concerned about the definition of the concept.

As shown in Table 3 digital HRM is conceptualized in different ways with different terminologies. This study has identified that digital HRM is not a new concept. It is a new terminology for the existing concept of e-HRM. All the existing studies referred to the previously developed understanding of e-HRM. Digital HRM is the advanced concept of web-based HRM (Strohmeier, 2020). It can be said that Electronic or Digital HRM is the digital transformation of HR functions and processes. Digital transformation is the process through which digital technologies merge and replace manual processes by focusing on business models to create significant value for stakeholders (Kraus et al., 2021). It aims to make an entity efficient through the adoption of information technology. As shown in Figure 8, this study extracted three themes from the key definitions. The first theme is the Human resource information system, which is a computer-based program. The second theme is the use of internet–based technology. The conceptual development of Digital HRM concept has shown in Figure 8.

In summary, digital HRM can be seen as the digital transformation of HR functions, leveraging information technology to enhance efficiency and create value. The study underscores that while the terminology may have evolved, the underlying principles and objectives remain consistent with those of e-HRM, focusing on the integration and optimization of HR processes through digital technologies. Based on the extracted themes present study defined digital HRM as *Digital HRM is the process of incorporating advanced information technology in existing processes to efficiently manage HR activities.* Digital HRM is the progress from web/technology-based HRM or HRIS (HR information system) concepts (Strohmeier, 2020). It has been clear that the concept of digital HRM has been used in a 'proliferating manner' by the authors who interchangeably used it with HRIS. It is a well-known phenomenon that has been denoted with different terminologies across time.

Reflexivity

Across the different conceptualizations of digital HRM with different terms presented in Tables 1,2 and 3 there are two commonalities a) efficient management and utilization of HR and related information and b) incorporation of advanced technology. The concept of digital HRM kept evolving during the

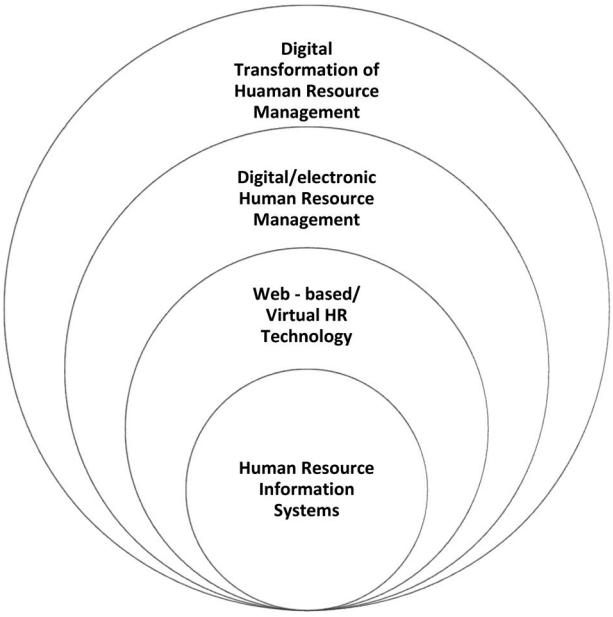

Figure 8. Conceptual development of Digital/Electronic Human Resource Management.

time of technology. Researchers put their efforts into advancing the terminology and extending the definition by incorporating new HR concepts, along with the technology. Across all the different key definitions, authors referred to Electronic and digital HRM as the usage, contribution, or incorporation of technology in terms of the internet, smart software, and application to perform HR activities or with the help of these tools efficiently and effectively manage HRM. Based on the analysis this study argues that HRIS is a computer-based program, it is not a process or phenomenon. Whereas e-HRM or digital HRM is the process of utilizing technology in the domain of HRM. Therefore, HRIS cannot be interchangeably used with e-HRM. In the aforementioned analysis, this study claimed that e-HRM and d-HRM are the same terms and support the interchangeable application of these terms. However, this study presented a separate definition for both concepts. From the literature search, it has been identified that e-HRM is interchangeably used with HRIS and d-HRM only. Other than these two concepts no other related concept of e-HRM has been identified. Whereas for the concept of d-HRM other than HRIS and e-HRM other related concepts were also identified including digitalization, digital transformation, and digitization. Studies that adopted these terms, did not claim these to be separate terms (Al-Dmour et al., 2017; Erdoğmuş & Esen, 2011; Masum et al., 2018; Panos & Bellou, 2016). Authors referred to digitalization and digital transformation as the d-HRM. The term digitization is not widely operationalized term; very few studies used this term but did not bother to explain it (Konovalova et al., 2021; Rana, 2019). Based on the existing developed understanding of e-HRM this study defines it separately. Whereas, in the case of d-HRM, this study defined it in the same manner but also attempted to explain the related term. And simultaneously incorporated these terms in the definition to clarify the ambiguity in the concept.

Conclusion

Based on the discussion and aforementioned arguments, it can be concluded that HRIS is misunderstood as a novel concept. HRIS incorrectly linked with and interchangeably used with well–established advanced digital HRM concepts. Whereas it is a computer-based program that helps HR managers efficiently perform HR activities. Another misunderstanding developed in literature is confusion about the concept of Electronic HRM and Digital HRM either these are interchangeable concepts or distinct. In literature, some authors defined these concepts interchangeably (Strohmeier, 2020; Thite, 2018). Some researchers defined electronic HRM as a distinct concept (Berber et al., 2018; Florkowski, 2018). The findings of this literature synthesis support the studies of Strohmeier (2020) and Thite (2018). It

can be said that HRIS is not a concept or process it is a computer-based system utilized to manage HR-related data and decision-making. Whereas electronic HRM and digital HRM are the same concepts. It is the process of utilizing HRIS and web-based technology. The purpose of both is to efficiently manage the HR data.

Limitation and future direction

The data of the last decade and particularly of the last five years have shown increase in the number of publications in the domain of digital HRM. A large number of publications are based on subjective descriptions of the DHRM concept and challenges. With the help of the SotA review present study has develop a clear understanding of the DHRM definition and its interchangeable terms. But this study has not addressed the second challenge of digital HRM research i.e. the practical challenges of DHRM. These challenges are unknown, few studies opt to indicate the present micro-level challenges such as resistance toward change, Alignment of HR process concerning e-business challenges, insufficient financial support, lack of managerial support, and technical and developmental issues (Alnadawi et al., 2022; Hernandez & Scotti, 2007; O. E. D. A. & S. O., 2015). Some studies also highlighted the macro level challenges that include, the development of appropriate employee capabilities, the rapid rise of platform capitalism, expansion of the intangible economy, globalization of the labor market, increase in involuntary different employment, impact on performance, digital ethics (Dabić et al., 2023; Zhang & Chen, 2023). These authors described these intraorganizational challenges based on their subjective understanding and experience. The description of these challenges is a mere description of early literature that explained the initial challenges organizations faced while implementing digital HR practices. The aforementioned studies lack the support of empirical shreds of evidence explaining the validity of these challenges in the present decade. This study calls for future research based on empirical evidence and field work. It will help researchers and practitioners to better understand the digital HRM related challenges and explore the solution. Recent studies argued that financial literacy, financial self-control and entrepreneurial orientation are important derivers of firm's performance (Mawad et al., 2022; Saleh & Athari, 2023). These factors may also influence the digitalization of human resource management. Future studies may consider these factors to examine the impact of adoption of digital HR practices.

Along with conceptual clarification, extensive research has highlighted the bright side of digitalization and emphasized on adoption of digitalized HRM (Bondarouk & Ruël, 2009; Manzoor & Sohaib, 2021; Prikshat et al., 2023; Thite, 2018). Digitalization of HRM has eased the many complicated processes and provided an opportunity to take their ordinary functions to a new level. Meanwhile, the advancement of digital technologies also has negative implications (Park, 2021). Despite the conspicuous benefit of the utilization of digital HRM, it is revealing the dark sides with severe impact at the individual, organizational, and societal levels (Turel et al., 2021; Vaghefi et al., 2017). At the individual level, some notable outcomes include technostress (Cadieux et al., 2021), negative health impact, security and privacy concerns, and online deviant behavior (Barlybaev et al., 2021; Blix & Levay, 2018; Palumbo & Cavallone, 2022). Similarly, studies on the negative side of digitalization at the organizational level identified issues related to employees' well–being, deviant workplace behavior (Christensen et al., 2020; Turel, 2017), tough competition due to imitable technology, rapid continuous restructuring of human resource functions, strategic use of digitalization (Bondarouk & Ruël, 2009; Fregnan et al., 2020). Likewise, research on the negative aspects of the digitalization of the economy highlighted the bulbous issue of loss or displacement of employment (Acemoglu & Restrepo, 2019; Badiuzzaman & Rafiquzzaman, 2020). Having such serious concerns raises a big question about the emphasis on digitalization or technology adoption. Future researchers should explore the negative impact of digital HRM on organizational and societal levels and suggest prospective solutions.

Note

1. Socio-technical refer as interrelationship between human and machine to foster the social and technical work conditions (Ropohl, 1999).

Disclosure statement

The Author(s) declare(s) that there is no conflict of interest.

Ethical statements

This article does not contain any studies with human participants performed by any of the authors.

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Data availability statement

The data used to carry out this study are available from the corresponding author upon request.

References

Acemoglu, D., & Restrepo, P. (2019). Automation and new tasks: How technology displaces and reinstates labor. *Journal of Economic Perspectives*, *33*(2), 3–30. https://doi.org/10.1257/jep.33.2.3

Al-Dmour, R. H. A., Ra', N. A., Masa', e., Deh, N. A., & Obeidat, B. Y. (2017). Factors influencing the adoption and implementation of HRIS applications: Are they similar? *International Journal of Business Innovation and Research*, *14*(2), 139–167. https://doi.org/10.1504/IJBIR.2017.086276

Alnadawi, A. A., Abdulrahman, M., & Omran, F. (2022). Opportunities and challenges of applying electronic human resources management in business organizations an applied study in the telecommunications sector, Jordan. In S. G. Yaseen (Ed.), *Digital economy, business analytics, and big data analytics applications* (pp. 223–235). Springer International Publishing. https://doi.org/10.1007/978-3-031-05258-3_19

Anthony, W. P. (1977). Get to know your employees – The human resource information system. Personnel *J*.

Badiuzzaman, M., & Rafiquzzaman, M. (2020). Automation and robotics: A review of potential threat on unskilled and lower skilled labour unemployment in highly populated countries. *International Business Management*, *14*(1), 16–24.

Barlybaev, A., Ishnazarova, Z., & Sitnova, I. (2021). Quality of life of the population: The impact of digitalization. *E3S Web of Conferences*, *295*, 34. https://doi.org/10.1051/e3sconf/202129501034

Barry, E. S., Merkebu, J., & Varpio, L. (2022). State-of-the-art literature review methodology: A six-step approach for knowledge synthesis. *Perspectives on Medical Education*, *11*(5), 281–288. https://doi.org/10.1007/s40037-022-00725-9

Berber, N., Đorđević, B., & Milanović, S. (2018). Electronic human resource management (e-HRM): A new concept for digital age. *Strategic Management*, *23*(2), 22–32. https://doi.org/10.5937/StraMan1802022B

Blix, M., & Levay, C. (2018). Digitalization and health care. *Eso Expertgrupp*, *44*, 13–35.

Bondarouk, T., Parry, E., & Furtmueller, E. (2017). Electronic HRM: Four decades of research on adoption and consequences. *The International Journal of Human Resource Management*, *28*(1), 98–131. https://doi.org/10.1080/09585192.2016.1245672

Bondarouk, T. V., & Ruël, H. J. M. (2009). Electronic human resource management: Challenges in the digital era. *The International Journal of Human Resource Management*, *20*(3), 505–514. https://doi.org/10.1080/09585190802707235

Broderick, R., & Boudreau, J. W. (1992). Human resource management, information technology, and the competitive edge. *Academy of Management Perspectives*, *6*(2), 7–17. https://doi.org/10.5465/ame.1992.4274391

Cadieux, N., Fournier, P.-L., Cadieux, J., & Gingues, M. (2021). New Techno-stressors among knowledge professionals: The contribution of artificial intelligence and websites that misinform clients. *International Journal of Electronic Commerce*, *25*(2), 136–153. https://doi.org/10.1080/10864415.2021.1887695

Chadegani, A. A., Salehi, H., Yunus, M. M., Farhadi, H., Fooladi, M., Farhadi, M., & Ebrahim, N. A. (2013). A comparison between two main academic literature collections: Web of science and scopus databases. *Asian Social Science*, *9*(5), 18. https://doi.org/10.5539/ass.v9n5p18

Christensen, J. O., Finne, L. B., Garde, A. H., Nielsen, M. B., Sørensen, K., & Vleeshouwes, J. (2020). The influence of digitalization and new technologies on psychosocial work environment and employee health: A literature review. *STAMI-Rapport*.

da Silva, L. B. P., Soltovski, R., Pontes, J., Treinta, F. T., Leitão, P., Mosconi, E., de Resende, L. M. M., & Yoshino, R. T. (2022). Human resources management 4.0: Literature review and trends. *Computers & Industrial Engineering*, *168*, 108111. https://doi.org/10.1016/j.cie.2022.108111

Dabić, M., Maley, J. F., Švarc, J., & Poček, J. (2023). Future of digital work: Challenges for sustainable human resources management. *Journal of Innovation & Knowledge*, *8*(2), 100353. https://doi.org/10.1016/j.jik.2023.100353

DeSanctis, G. (1986). Human resource information systems: A current assessment. *MIS Quarterly*, *10*(1), 15–27. https://doi.org/10.2307/248875

Dilu, E., Gebreslassie, M., & Kebede, M. (2017). Human Resource Information System implementation readiness in the Ethiopian health sector: A cross-sectional study. *Human Resources for Health*, *15*(1), 85. https://doi.org/10.1186/s12960-017-0259-3

Donthu, N., Kumar, S., & Pattnaik, D. (2020). Forty-five years of Journal of Business Research: A bibliometric analysis. *Journal of Business Research*, *109*, 1–14. https://doi.org/10.1016/j.jbusres.2019.10.039

Dosi, G., & Nelson, R. R. (2010). Chapter 3 – Technical change and industrial dynamics as evolutionary processes. In B. H. Hall & N. Rosenberg (Eds.), *Handbook of the economics of innovation* (Vol. 1, pp. 51–127). https://doi.org/10.1016/S0169-7218(10)01003-8

Ebert, C., & Duarte, C. H. (2018). Digital transformation. *IEEE Software*, *35*(4), 16–21. https://doi.org/10.1109/MS.2018.2801537

Erdoğmuş, N., & Esen, M. (2011). An investigation of the effects of technology readiness on technology acceptance in e-HRM. *Procedia – Social and Behavioral Sciences*, *24*, 487–495. https://doi.org/10.1016/j.sbspro.2011.09.131

Florkowski, G. W. (2018). HR technology systems: An evidence-based approach to construct measurement. In M. R. Buckley, A. R. Wheeler, & J. R. B. Halbesleben (Eds.), *Research in personnel and human resources management* (Vol. 36, pp. 197–239). Emerald Publishing Limited. https://doi.org/10.1108/S0742-730120180000036006

Fosso Wamba, S., Akter, S., Edwards, A., Chopin, G., & Gnanzou, D. (2015). How 'big data' can make big impact: Findings from a systematic review and a longitudinal case study. *International Journal of Production Economics*, *165*, 234–246. https://doi.org/10.1016/j.ijpe.2014.12.031

Fregnan, E., Ivaldi, S., & Scaratti, G. (2020). HRM 4.0 and new managerial competences profile: The COMAU Case. *Frontiers in Psychology*, *11*, 578251. https://doi.org/10.3389/fpsyg.2020.578251

Grant, M. J., & Booth, A. (2009). A typology of reviews: An analysis of 14 review types and associated methodologies. *Health Information & Libraries Journal*, *26*(2), 91–108. https://doi.org/10.1111/j.1471-1842.2009.00848.x

He, W., Zhang, Z. (Justin), & Li, W. (2021). Information technology solutions, challenges, and suggestions for tackling the COVID-19 pandemic. *International Journal of Information Management*, *57*, 102287. https://doi.org/10.1016/j.ijinfomgt.2020.102287

Hernandez, S., & Scotti, N. (2007). The challenge of the change management in an information and management systems unit: What strategy to manage the human resources? *International Journal of Human Resources Development and Management*, *7*(1), 83–103. https://doi.org/10.1504/IJHRDM.2007.012287

Hyde, A. C., & Shafritz, J. M. (1977). HRIS: Introduction to tomorrow's system for managing human resources. *Public Personnel Management*, *6*(2), 70–77. https://doi.org/10.1177/009102607700600201

Imamov, M., & Semenikhina, N. (2021). The impact of the digital revolution on the global economy. *Linguistics and Culture Review*, *5*(S4), 968–987. https://doi.org/10.21744/lingcure.v5nS4.1775

Jani, A., Muduli, A., & Kishore, K. (2021). Human resource transformation in India: Examining the role digital human resource technology and human resource role. *International Journal of Organizational Analysis*, *31*(4), 959–972. https://doi.org/10.1108/IJOA-08-2021-2886

Jesser, P. (1990). Human resource information systems: Development and application by Michael J. Kavanagh, Hal G. Gueutal, and Scott I. Tannenbaum (1990). *Asia Pacific Journal of Human Resources*, *28*(4), 144–144. https://doi.org/10.1177/103841119002800422

Konovalova, V. G., Aghgashyan, R. V., & Galazova, S. S. (2021). Perspectives and restraining factors of HR analytics in the conditions of digitization of human resources management. In E. G. Popkova, V. N. Ostrovskaya, & A. V. Bogoviz (Eds.), *Socio-economic Systems: Paradigms for the future* (pp. 1015–1024). Springer International Publishing. https://doi.org/10.1007/978-3-030-56433-9_106

Kraus, S., Jones, P., Kailer, N., Weinmann, A., Chaparro-Banegas, N., & Roig-Tierno, N. (2021). Digital transformation: An overview of the current state of the art of research. *SAGE Open*, *11*(3), 215824402110475. https://doi.org/10.1177/21582440211047576

Lederer, A. L. (1984a). Information technology: 1. Planning and developing a human resources information system. *Personnel*, *61*(3), 14–27.

Lederer, A. L. (1984b). Planning and developing a human resource information system. The logic of a step-by-step approach. *The Personnel Administrator*, *29*(8), 27–39.

Lengnick-Hall, M. L., & Moritz, S. (2003). The impact of e-HR on the human resource management function. *Journal of Labor Research*, *24*(3), 365–379. 10.1007/s12122-003-1001-6

Lepak, D. P., & Snell, S. A. (1998). Virtual HR: Strategic human resource management in the 21st century. *Human Resource Management Review*, *8*(3), 215–234. 10.1016/S1053-4822(98)90003-1

Luftman, J. N., Lewis, P. R., & Oldach, S. H. (1993). Transforming the enterprise: The alignment of business and information technology strategies. *IBM Systems Journal*, *32*(1), 198–221. https://doi.org/10.1147/sj.321.0198

Manzoor, A., & Sohaib, M. (2021). Exploring the critical functions of electronic human resource management in banking Sector of Pakistan: Exploring the critical functions of electronic human resource management. *Journal of Social Sciences*, *12*(1), 49.

Marler, J. H., & Parry, E. (2016). Human resource management, strategic involvement and e-HRM technology. *The International Journal of Human Resource Management*, *27*(19), 2233–2253. https://doi.org/10.1080/09585192.2015.1091980

Masum, A. K. M., Beh, L. S., Azad, A. K., & Hoque, K. E. (2018). Intelligent human resource information system (i-HRIS): A holistic decision support framework for HR excellence. *International Arab Journal of Information Technology*, *15*(1), Article 1.

Mawad, J. L., Athari, S. A., Khalife, D., & Mawad, N. (2022). Examining the impact of financial literacy, financial self-control, and demographic determinants on individual financial performance and behavior: An insight from the lebanese crisis period. *Sustainability*, *14*(22), 15129. https://doi.org/10.3390/su142215129

Mitlacher, L. W. (2012). Appraisal and rewards systems for virtual project management teams and the challenges for human resource management. *International Journal of Networking and Virtual Organisations*, *10*(2), 153–168. https://doi.org/10.1504/IJNVO.2012.045732

Nikkel, D. (1985). HRIS implementation: A systematic approach. *Personnel*, *62*(2), 10–15.

O. E. D. A. & S. O. (2015). Implementing E-HRM system in developing countries: Challenges and prospects. *International Journal of Applied Information Systems*, *9*(8), 38–41. https://doi.org/10.5120/ijais2015451452

Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., … Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *International Journal of Surgery*, *88*, 105906. https://doi.org/10.1016/j.ijsu.2021.105906

Palumbo, R., & Cavallone, M. (2022). Is work digitalization without risk? Unveiling the psycho-social hazards of digitalization in the education and healthcare workplace. *Technology Analysis & Strategic Management*, *36*(6), 1136–1149. https://doi.org/10.1080/09537325.2022.2075338

Panos, S., & Bellou, V. (2016). Maximizing e-HRM outcomes: A moderated mediation path. *Management Decision*, *54*(5), 1088–1109. https://doi.org/10.1108/MD-07-2015-0269

Park, H. E. (2021). *The Dark side of digitalization: Three case studies of the negative implications of information technology* [PhD Thesis]. https://ses.library.usyd.edu.au/handle/2123/26410

Parviainen, P., Tihinen, M., Kääriäinen, J., & Teppola, S. (2017). Tackling the digitalization challenge: How to benefit from digitalization in practice. *International Journal of Information Systems and Project Management*, *5*(1), 63–77. https://doi.org/10.12821/ijispm050104

Prikshat, V., Malik, A., & Budhwar, P. (2023). AI-augmented HRM: Antecedents, assimilation and multilevel consequences. *Human Resource Management Review*, *33*(1), 100860. https://doi.org/10.1016/j.hrmr.2021.100860

Rana, D. T. (2019). *Digitization of human resource practices – An emerging trend* (SSRN Scholarly Paper 3323740). https://doi.org/10.2139/ssrn.3323740

Ropohl, G. (1999). Philosophy of socio-technical systems. *Society for Philosophy and Technology Quarterly Electronic Journal*, *4*(3), 186–194. https://doi.org/10.5840/techne19994311

Ruël, H., & Bondarouk, T. (2014). e-HRM research and practice: Facing the challenges ahead. In F. J. Martínez-López (Ed.), *Handbook of strategic e-business management* (pp. 633–653). Springer. https://doi.org/10.1007/978-3-642-39747-9_26

S, D. S., & Aggarwal, V. (2017). Digital human resource management. *Gyan Management*, *11*(2), Article 2. https://acspublisher.com/journals/index.php/gmj/article/view/521

Saleh, A. M., & Athari, S. A. (2023). Examining the impact of entrepreneurial orientation on new venture performance in the emerging economy of Lebanon: A moderated mediation analysis. *Sustainability*, *15*(15), 11982. https://doi.org/10.3390/su151511982

Sima, V., Gheorghe, I. G., Subić, J., & Nancu, D. (2020). Influences of the Industry 4.0 Revolution on the human capital development and consumer behavior: A systematic review. *Sustainability*, *12*(10), 4035. https://doi.org/10.3390/su12104035

Simon, S. H. (1983). The HRIS: What capabilities must it have. *Personnel*, *60*(5), 36–49.
Snyder, H. (2019). Literature review as a research methodology: An overview and guidelines. *Journal of Business Research*, *104*, 333–339. https://doi.org/10.1016/j.jbusres.2019.07.039
Strohmeier, S. (2007). Research in e-HRM: Review and implications. *Human Resource Management Review*, *17*(1), 19–37. https://doi.org/10.1016/j.hrmr.2006.11.002
Strohmeier, S. (2020). Digital human resource management: A conceptual clarification. *German Journal of Human Resource Management: Zeitschrift für Personalforschung*, *34*(3), 345–365. https://doi.org/10.1177/2397002220921131
Thite, M. (2018). Electronic/digital HRM: A primer. In *E-HRM*. Routledge.
Thite, M. (2022). Digital human resource development: Where are we? Where should we go and how do we go there? *Human Resource Development International*, *25*(1), 87–103. https://doi.org/10.1080/13678868.2020.1842982
Turel, O. (2017). Organizational deviance via social networking site use: The roles of inhibition, stress and sex differences. *Personality and Individual Differences*, *119*, 311–316. https://doi.org/10.1016/j.paid.2017.08.002
Turel, O., Qahri-Saremi, H., & Vaghefi, I. (2021). Special issue: Dark sides of digitalization. *International Journal of Electronic Commerce*, *25*(2), 127–135. https://doi.org/10.1080/10864415.2021.1887694
Vaghefi, I., Lapointe, L., & Boudreau-Pinsonneault, C. (2017). A typology of user liability to IT addiction. *Information Systems Journal*, *27*(2), 125–169. https://doi.org/10.1111/isj.12098
van Eck, N. J., & Waltman, L. (2011). *Text mining and visualization using VOSviewer* [Computer software]. http://arxiv.org/abs/1109.2058
Vial, G. (2021). Understanding digital transformation: A review and a research agenda. In *Managing Digital Transformation* (1st ed., pp. 13–66).
Von Glinow, M. A., & Teagarden, M. B. (1988). The transfer of human resource management technology in Sino-U.S. cooperative ventures: Problems and solutions. *Human Resource Management*, *27*(2), 201–229. https://doi.org/10.1002/hrm.3930270206
Walker, A. J. (1982). *HRIS development: A project team guide to building an effective personnel information system*. Van Nostrand Reinhold.
Wen, X. (2013 *E-HRM in Chinese Organizations: Managing human resources with information technology in digital age* [Paper presentation]. In 2013 International Conference on Computational and Information Sciences, 545–548. https://doi.org/10.1109/ICCIS.2013.151
White, M. (2012). Digital workplaces: Vision and reality. *Business Information Review*, *29*(4), 205–214. https://doi.org/10.1177/0266382112470412
Wilkens, P. L. (1971). *An analysis of the attitudinal and behavioral impact of a human resource information system* [PhD Thesis]. The Ohio State University.
Wirtky, T., Laumer, S., Eckhardt, A., & Weitzel, T. (2016). On the untapped value of e-HRM: A literature review. *Communications of the Association for Information Systems*, *38*(1), 20–83. https://doi.org/10.17705/1CAIS.03802
Zavyalova, E., Sokolov, D., Kucherov, D., & Lisovskaya, A. (2022). The digitalization of human resource management: Present and future. *Foresight and Sti Governance*, *16*(2), 42–51. https://doi.org/10.17323/2500-2597.2022.2.42.51
Zhang, J., & Chen, Z. (2023). Exploring human resource management digital transformation in the digital age. *Journal of the Knowledge Economy*, *15*(1), 1482–1498. https://doi.org/10.1007/s13132-023-01214-y
Ziebell, R.-C., Albors-Garrigos, J., Schoeneberg, K.-P., & Marin, M. R. P. (2019). e-HRM in a cloud environment: Implementation and its adoption: A literature review. *International Journal of Human Capital and Information Technology Professionals*, *10*(4), 16–40. https://doi.org/10.4018/IJHCITP.2019100102